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Studierichting: Informatica
Oefeningenreeks 5K nr. 1.3.

$$r = 2 - 2\cos\theta$$

$$r' = 2\sin\theta$$

$$L = \int_0^{2\pi} \sqrt{(2-2\cos(\theta))^2 + (2\sin(\theta)^2)}d\theta$$

$$= \int_0^{2\pi} \sqrt{4(1-\cos(\theta))^2 + 4\sin(\theta)^2}d\theta$$

$$= \int_0^{2\pi} 2\sqrt{(1-\cos(\theta))^2 + \sin(\theta)^2}d\theta$$

$$= \int_0^{2\pi} 2\sqrt{1 - 2\cos(\theta) + \cos^2(\theta) + \sin^2(\theta)}d\theta$$

$$= \int_0^{2\pi} 2\sqrt{2-2\cos(\theta)}d\theta$$

$$= \int_0^{2\pi} 2 \cdot \sqrt{2} \cdot \sqrt{2}\sin\frac{\theta}{2}d\theta$$

$$= \int_0^{2\pi} 4\sin\frac{\theta}{2}d\theta$$

$$= 8[-\cos(u)]_0^{\pi}$$

$$= 16$$

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References